LAGUNA STATE POLYTECHNIC UNIVERSITY

Professional Installation Manual

Ubuntu Server LTS | Reproducible Deployment Procedure

Alay, Riz-Anne V.
BSIT-4E

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Mr. Penaredondo

1. Purpose, Scope, and Audience

This manual provides an end-to-end procedure for deploying Ubuntu Server LTS as a virtual machine. It is written for system administrators who must reproduce, verify, secure, and hand over a baseline server deployment.

- Virtual machine creation and ISO attachment.
- Interactive installation using DHCP and guided whole-disk storage.
- Hostname server01, non-root administrative account, and OpenSSH.
- Post-installation update, network, and service checks.
- Evidence collection, troubleshooting, and operational handover.

2. Change and Safety Controls

Control	Required action
Authorization	Confirm the assigned Week 3 laboratory scope.
Recovery	Establish a snapshot and independent backup policy.
Media	Use an official Ubuntu Server LTS ISO.
Credentials	Use a strong unique password stored securely.
Evidence	Crop screenshots and redact sensitive data.

3. Pre-deployment Checklist

Check	Status / value
Virtualization platform installed	[/]
Ubuntu Server LTS ISO acquired	[/]
4 GB RAM available	[/]
2 virtual CPUs available	[/]
40 GB virtual disk available	[/]
Network mode approved	[/]
Hostname server01 approved	[/]
Administrative username selected	[/]

4. Installation Procedure

1. Create the virtual machine

Create Ubuntu-Server-Week03 in VirtualBox or VMware. Assign 4 GB RAM, 2 virtual processors, a 40 GB virtual disk, NAT networking, and attach the Ubuntu Server LTS ISO.

2. Start the installer

Boot from the ISO. Select English and the appropriate keyboard layout, such as English (US).

3. Configure networking

Accept DHCP unless instructed otherwise. Record the assigned IP address. Leave the proxy blank unless required and use a reachable Ubuntu mirror.

4. Configure storage

Select Guided - use an entire disk. Confirm that the selected device is the 40 GB virtual disk. Record the partition scheme, file system, and disk size.

5. Create the server identity

Set hostname to server01. Enter your full name, choose a non-root username, and use a strong unique password.

6. Enable SSH

Select Install OpenSSH server. Skip extra packages unless instructed.

7. Complete installation

Wait for completion, reboot, and remove the ISO if prompted.

Installation acceptance gate

Proceed only when the server restarts from the virtual disk and presents a login prompt. If the installer restarts, power off, detach the ISO, and boot again.

5. Initial Server Configuration

After first login, capture baseline information and install updates.

```
hostname
ip addr
sudo apt update
sudo apt upgrade -y
systemctl status ssh
```

Optional hardening baseline

- Use SSH public-key authentication after testing it.
- Restrict SSH to approved accounts and networks.
- Configure the host firewall for required services only.
- Apply security updates on an approved schedule.
- Use least privilege and sudo rather than routine root login.
- Maintain logs, time synchronization, monitoring, and backups.

6. Verification and Acceptance

Test	Command / action	Pass criterion
Login	Console login	Administrative user reaches shell
Hostname	hostname	server01
IP address	ip addr	Expected assigned IPv4
Connectivity	ping -c 4 google.com	Four replies or successful equivalent test
Updates	apt update / apt upgrade	No unresolved errors
SSH	systemctl status ssh	active (running)

7. Troubleshooting Guide

Symptom	Likely cause	Corrective action
VM does not boot installer	ISO or boot order problem	Attach ISO and place optical media first for initial boot.
Installer repeats after reboot	ISO remains attached	Detach ISO and boot from virtual disk.
No IP address	DHCP unavailable or adapter disconnected	Check VM network adapter; renew DHCP or follow approved static-IP procedure.
DNS ping fails	DNS or network issue	Test gateway/IP connectivity and inspect resolver status.
APT update fails	Network, mirror, time, or lock issue	Confirm connectivity and time; wait for other package processes; verify mirror.
SSH inactive	Package missing or	Install openssh-server; inspect systemctl

	service error	status ssh and journalctl -u ssh.
Remote SSH timeout	Wrong IP, NAT, firewall, or service	Verify IP/service; use approved bridged mode or NAT forwarding; check firewall.
Login rejected	Wrong account/password/layout	Confirm username and keyboard layout; use approved recovery.

8. Operational Best Practices

- Document changes with date, reason, owner, and validation result.
- Patch regularly and test critical changes.
- Use key-based SSH, least privilege, firewall policy, log review, and backups.
- Treat snapshots as short-term recovery points, not durable backups.
- Remove unused services and expose only required ports.
- Monitor capacity, availability, authentication events, and restoration tests.